EXPLAINABLE AI FOR CANCER SUBTYPE PREDICTION FROM GENE EXPRESSION

Project Specification, Proposed Methodology and Risk Management Plan

Project focus: Breast cancer molecular subtype prediction

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This document defines the agreed scope, proposed technical approach, intended software functionality and principal project risks. Planned experiments and deliverables are distinguished from completed work.

1. Problem Statement, Aim and Objectives

1.1 Problem Statement

Breast cancer is a heterogeneous disease comprising molecular subtypes with distinct gene-expression patterns. Identifying these subtypes is important for understanding tumor biology and can have clinical relevance. Established gene-expression-based approaches, including PAM50, provide a foundation for molecular subtype classification. However, developing machine-learning classifiers from high-dimensional RNA-seq-derived expression profiles presents challenges involving feature selection, model generalization and interpretability.

A predicted subtype alone does not reveal which genes influenced the classifier, nor does it establish the biological significance of those genes. This project addresses these challenges by developing and evaluating an explainable AI application for breast cancer molecular subtype prediction, integrating gene-level model explanations with information supported by scientific literature.

1.2 Project Aim

To develop and evaluate a working, web-based machine-learning application that predicts breast cancer molecular subtypes from processed RNA-seq-derived gene-expression profiles and presents understandable, scientifically qualified explanations of its predictions.

1.3 Objectives

• Prepare and validate the GSE96058 gene-expression dataset and its PAM50 subtype annotations.

• Investigate the available PAM50 gene signature and compare it with data-driven feature subsets.

• Train and compare Logistic Regression, Support Vector Machine, Random Forest and XGBoost classifiers.

• Evaluate performance on unseen samples using macro-F1 as the primary model-selection metric and per-subtype measures for detailed assessment.

• Use SHAP to explain individual predictions and summarize influential genes across evaluated samples.

• Link important genes to curated, verifiable scientific evidence without presenting model attributions as biological causation.

• Integrate the selected model into a functional Streamlit application supporting compatible single-sample and batch inputs, interpretable outputs and downloadable PDF reports.

2. Project Scope and Limitations

2.1 Included Scope

The project is limited to **breast cancer molecular subtype prediction**. The primary development dataset is NCBI GEO GSE96058, which contains RNA-seq-derived expression data and PAM50 subtype annotations. The initial experimental setting will investigate all five annotated subtypes: Luminal A, Luminal B, HER2-enriched, Basal-like and Normal-like. The reliability of Normal-like classification will be specifically examined before finalizing the class configuration.

The application will allow researchers to submit one or multiple compatible, processed gene-expression profiles; obtain a subtype prediction and class scores; review SHAP-based gene contributions and curated biological context; and export a research-use PDF report. The model and required preprocessing will be developed offline and incorporated into the application after evaluation.

2.2 Exclusions and Limitations

• **Raw sequencing processing:** the application will not convert raw FASTQ files into gene-expression measurements. It accepts previously processed profiles and applies only the model-compatible preprocessing required for inference.

• **Other cancers and prognosis:** prediction of other cancer types, cancer-type identification, survival, recurrence and treatment response are outside the committed scope.

• **Clinical use:** the application is a research and educational prototype; it does not detect cancer, replace diagnostic assays, recommend treatments or claim clinical validation.

• **Input generalizability:** only profiles matching documented gene identifiers, expression representation and required feature schema will be accepted.

• **Interpretation:** SHAP describes how a model makes predictions; it does not demonstrate that a gene causes a subtype or confirms a clinical mechanism.

2.3 Potential Future Extensions

Additional cancer-specific models, automatic cancer-type identification and prognostic modeling may be investigated in subsequent versions, subject to suitable datasets and independent validation. These are not current deliverables.

3. System Requirements

Functional requirements specify what the application must do. Non-functional requirements specify the conditions under which it should perform. Requirements have been consolidated to avoid repetition while retaining essential coverage.

3.1 Functional Requirements

ID	Function	Requirement
FR-01	Sample input	Accept one or multiple processed breast cancer gene-expression profiles in a documented supported file format.
FR-02	Input validation	Check gene identifiers, required features, numeric values, missing or duplicate entries and compatibility with the supported expression representation; clearly reject invalid inputs.
FR-03	Consistent preprocessing	Apply the saved, training-compatible preprocessing steps to accepted expression profiles.
FR-04	Subtype prediction	Automatically use the selected trained classifier to return a molecular subtype for each valid sample; no cancer-type or algorithm selection is required.
FR-05	Prediction scores	Display class scores and communicate their interpretation and any evaluated uncertainty limitation; do not label uncalibrated scores as probabilities.
FR-06	Sample-specific explanation	Generate SHAP visualizations identifying influential genes and the direction of their contribution to the explained model output.
FR-07	Global model explanation	Present a summary of influential genes across a defined evaluation sample set, clearly distinguished from patient-specific explanations.
FR-08	Biological interpretation	Display available verified information about influential genes and breast cancer, with supporting clickable references; identify unavailable or uncertain evidence.
FR-09	Results interface	Present predictions, scores, explanations and biological context through an understandable Streamlit interface; preserve sample identifiers for batch results.
FR-10	PDF reporting	Allow export of a readable report containing predictions, class scores, important genes, explanation visualizations, available biological context, sources and model details.
FR-11	Clear feedback	Explain unsupported files, failed validations and prediction errors without showing fabricated results.
FR-12	Research-use disclosure	Show a clear research-use-only notice in the application and exported reports.

3.2 Non-Functional Requirements

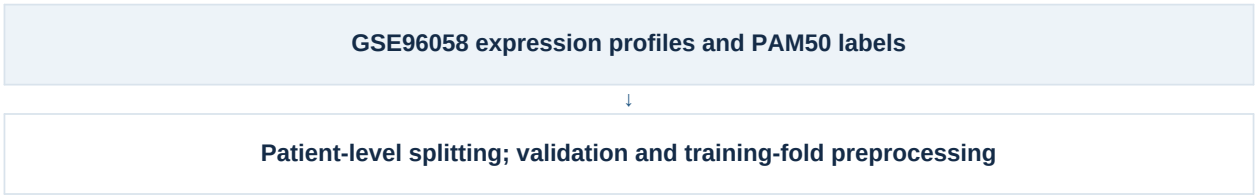
ID	Quality	Verification criterion
NFR-01	Scientific evaluation	Report macro-F1, accuracy, per-class precision/recall/F1 and confusion matrices on held-out data; document cross-validation results.
NFR-02	Leakage prevention	Separate samples at patient level, keeping technical replicates together; fit data-driven feature selection and preprocessing inside training folds only.
NFR-03	Reproducibility	Record data versions, code and software dependencies, random seeds, model parameters, preprocessing and selected genes so experiments can be repeated.
NFR-04	Input integrity	Use a versioned feature schema and reject incompatible data rather than silently substituting or inventing missing expression measurements.
NFR-05	Explanation validity	Confirm that SHAP explanations correspond to the deployed model output and examine meaningful variation in gene-importance results.
NFR-06	Evidence integrity	Support biological statements with traceable verified sources and distinguish literature-supported associations from model explanations.
NFR-07	Usability	Enable the core upload-to-report workflow without programming knowledge, with readable labels, figures and error messages.
NFR-08	Performance	Benchmark single-sample and batch inference on documented hardware; provisional targets are 30 seconds for one sample and five minutes for 100 samples, including explanations.
NFR-09	Reliability	Handle rejected files and processing failures without crashing or producing misleading results; test typical and invalid input cases.
NFR-10	Privacy and security	Avoid requesting personal identifiers, do not retain uploaded profiles by default and treat files as data rather than executable content.
NFR-11	Maintainability	Separate validation, preprocessing, inference, SHAP, curated evidence and reporting modules, with documented setup and dependencies.
NFR-12	Compatibility	Test the core interface in current mainstream desktop browsers and document the supported execution environment.

Performance targets are design objectives, not measured results. The final report will state the achieved values and test environment.

4. Proposed System Architecture

The solution separates offline model development from user-facing inference. This separation keeps experimentation reproducible and avoids retraining a classifier whenever a user uploads a sample.

4.1 Development Workflow



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Feature-subset experiments and cross-validated model comparison

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Held-out evaluation and selection of the final classifier

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Export of versioned model, fitted preprocessing and input schema

4.2 Application Workflow

Upload processed expression profile(s)

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Validate file, gene identifiers and required expression representation

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Apply saved pipeline and generate subtype predictions

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Generate model-specific SHAP explanations; retrieve curated gene evidence

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Display results and export research-use PDF report

4.3 Component Responsibilities

Component	Responsibility
Data validation and preprocessing	Enforce the supported file schema and reproduce fitted transformations.
Classifier	Load the selected trained model and generate class labels and scores.
SHAP module	Produce class-specific feature contributions and model-level summaries.
Evidence module	Supply curated gene descriptions and linked references; flag evidence gaps.
Streamlit and reporting	Coordinate input, results display, user guidance and PDF exports.

5. Proposed Methodology

5.1 Dataset and Initial Exploration

The primary dataset is GSE96058 (SCAN-B). The current exploratory analysis notebook reports 3,273 primary tumor samples and 30,865 gene-expression features, with 136 technical replicates excluded. Its reported expression
The notebook reports log2(FPKM + 0.1) expression and all 50 PAM50 genes after resolving the historical ORC6L symbol to ORC6 (49 direct matches and one alias match). These are EDA findings, not model results.

We will inspect subtype frequencies, missing and non-finite values, duplicate identifiers, gene coverage and sample metadata. The relationship between technical replicate identifiers and patient identities will be checked before splitting.

5.2 Data Partitioning and Preprocessing

A provisional 80% development and 20% held-out test partition will be used, with patient-level separation and attention to subtype representation. Five-fold cross-validation is planned within development data where the available groups and class counts permit. Feature selection and any fitted scaling or imputation must occur within each training fold. The test set will remain untouched until the final evaluation.

5.3 Gene Selection Experiments

We will compare (a) models trained on the available PAM50-associated genes, (b) ANOVA-based subsets selected within training folds, and (c) sparse subsets identified using L1 or elastic-net regularization. Candidate subset sizes will be evaluated empirically. Models trained on PAM50 genes are **not** automatically equivalent to the original PAM50 centroid classifier; that distinction will be explicit in the experimental report.

5.4 Classifier Comparison

Regularized Logistic Regression, Support Vector Machine, Random Forest and XGBoost will be compared using consistent partitions and a defined hyperparameter-search procedure. Selection will prioritize cross-validated macro-F1 and will also consider subtype-specific performance, stability and operational suitability. The final model will be chosen from evidence rather than predetermined.

5.5 Evaluation and Subtype Reliability

The final evaluation will report accuracy, macro-F1, per-subtype precision/recall/F1 and confusion matrices. Particular attention will be paid to Normal-like samples, including class overlap and confusion with other labels. Any later decision to use four rather than five classes must be justified scientifically and disclosed consistently in the application and documentation. We will not remove a class solely to improve a headline score.

5.6 Explainability and Biological Context

SHAP explanations will be computed for the deployed model, using the appropriate class-specific output for each prediction. Model-specific gene contributions will be presented separately from verified biological descriptions. The project will examine the stability of important-gene rankings across suitable evaluation settings and will not interpret SHAP as causal evidence. A curated reference collection will support concise, traceable gene annotations.

5.7 Application Integration and Validation

The final inference package will include the trained classifier, preprocessing, required features, label mapping and version metadata. The Streamlit application will be tested with valid single and batch files, missing required genes, unsupported expression formats and malformed values. Successful tests will check prediction outputs, SHAP plots, evidence links, model metadata, sample identifiers and PDF exports.

6. Technology Stack and Expected Deliverables

6.1 Planned Technologies

Technology	Planned use
Python; Pandas; NumPy	Data preparation and application logic.
Scikit-learn	Model pipelines, feature selection, validation and standard classifiers.
XGBoost	Boosted-tree classification experiments.
SHAP	Model-level and sample-level gene contribution analysis.
Jupyter Notebook	Exploratory analysis and reproducible experiments.
Matplotlib; Seaborn	Scientific plots and evaluation figures.
Streamlit	Web interface for inference and explanation.
Git; GitHub	Source control, team collaboration and version history.
Python PDF library (to be selected)	Generation of downloadable prediction reports.

A dedicated database and external generative AI service are not required for the first version. Curated biological information and model artifacts may be stored as versioned local resources.

6.2 Final Project Deliverables

• Validated GSE96058 preparation pipeline and documented exploratory data analysis.

• Trained breast cancer subtype classifier with saved preprocessing, feature schema and evaluation results.

• Reproducible comparison of models and gene-selection methods, including SHAP analyses.

• Functional Streamlit application for compatible single and batch expression-profile input.

• Downloadable research-use PDF prediction reports with traceable biological references.

• Organized source-code repository, setup instructions, tests and final academic project documentation.

These are **planned deliverables**. Their inclusion does not imply that model training, application integration or independent validation has already been completed.

7. Risk Assessment and Mitigation Plan

The risk register prioritizes threats to scientific validity, interpretability, usability and completion. Likelihood and impact are preliminary qualitative assessments that the team should reassess as development progresses.

ID	Risk	Likelihood	Impact	Priority
R1	Data leakage produces overly optimistic scores	Medium	High	High
R2	Weak classification of some PAM50 subtypes, particularly Normal-like	High	High	High
R3	Uploaded expression data are incompatible with training inputs	High	High	High

ID	Risk	Likelihood	Impact	Priority
R4	SHAP findings or gene descriptions are misleading	Medium	High	High
R5	Project delays affect model development or application integration	Medium	High	High
R6	Predictions do not transfer reliably to other cohorts	High	Medium	High

7.1 Risk Mitigation Actions

R1 - Data leakage produces overly optimistic scores

Partition by patient before fitting learned transforms; keep replicates together; select genes and tune models only within training folds; evaluate once on the held-out test set.

R2 - Weak classification of some PAM50 subtypes, particularly Normal-like

Report per-class metrics and confusion patterns; examine subtype annotations and imbalance; compare modeling approaches; justify and document any scientifically supported change to the class configuration.

R3 - Uploaded expression data are incompatible with training inputs

Publish the required feature identifiers and expression representation; use explicit schema checks and training-compatible transformations; reject incompatible submissions with actionable messages.

R4 - SHAP findings or gene descriptions are misleading

Use SHAP with the selected model and correct output; verify attributions, assess important-gene stability and separate model associations from curated biological evidence; flag unsupported interpretations.

R5 - Project delays affect model development or application integration

Track milestone deliverables and integration checks in GitHub; secure a working end-to-end baseline early; develop the reporting/interface components against representative sample outputs while models are compared.

R6 - Predictions do not transfer reliably to other cohorts

Document GSE96058-specific preprocessing and study limitations; examine cohort compatibility, attempt a genuinely independent suitable evaluation dataset if feasible and avoid claims of external or clinical validity without testing.

7.2 Monitoring and Response

Risks will be revisited at project milestones and whenever a new issue affects dataset integrity, model evaluation, explanation quality or application readiness. The team will record changes in priority, mitigation actions and remaining issues in its project tracking workflow. High-priority threats to scientific validity should be addressed before performance claims or system demonstrations are finalized.

Document status: Consolidated project specification based on the agreed proposed scope and methodology. Experimental decisions and measured results will be updated as validation proceeds.